

Differential Geometry 5

Theory: Plane cubics and the Cremona involution

Plane cubics

Assume k is algebraically closed.

A *plane cubic* is a curve

$$C \subset \mathbb{P}^2$$

defined by a single homogeneous polynomial of degree 3:

$$C = Z(F), \quad F \in k[x, y, z] \text{ homogeneous of degree 3.}$$

An affine cubic in $\mathbb{A}^2$ given by $f(x, y)$ has projective closure obtained by homogenization:

$$F(x, y, z) = z^3 f\left(\frac{x}{z}, \frac{y}{z}\right).$$

Classification by reducibility and singularities

Every plane cubic falls into one of the following types.

Smooth cubic. If C is smooth, then it is irreducible and has genus

$$g(C) = 1.$$

After choosing a base point $O \in C$, it carries the structure of an elliptic curve via the chord–tangent law. A line meets C in 3 points counting multiplicity.

Singular irreducible cubic. If C is irreducible but singular, then it is rational:

$$g(C) = 0.$$

The typical singularities are a node or a cusp, and C admits a rational parametrization.

Reducible cubic. If F factors, then C is reducible. The possibilities are:

three lines (with multiplicities),

or

a line plus a conic (with multiplicities).

For example,

$$Z(xyz) = Z(x) \cup Z(y) \cup Z(z), \quad Z(z(xy - z^2)) = Z(z) \cup Z(xy - z^2).$$

Bezout's theorem for cubics

If $C = Z(F)$ and $D = Z(G)$ are plane curves in $\mathbb{P}^2$ with no common component, then

$$\sum_{p \in C \cap D} I_p(C, D) = \deg(F) \deg(G).$$

In particular, a line meets a cubic in

$$3$$

points counting multiplicity, and a conic meets a cubic in

$$6$$

points counting multiplicity.

Weierstrass form

Assume $\text{char}(k) \neq 2, 3$. Every smooth plane cubic is projectively equivalent to a Weierstrass model

$$y^2z = x^3 + axz^2 + bz^3,$$

with discriminant $\Delta \neq 0$. In this form, the group law has the explicit inversion

$$(x : y : z) \longmapsto (x : -y : z).$$

The standard Cremona involution

Define the rational map

$$\sigma : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \quad (x : y : z) \longmapsto (yz : zx : xy).$$

It is undefined exactly at the three base points

$$p_1 = (1 : 0 : 0), \quad p_2 = (0 : 1 : 0), \quad p_3 = (0 : 0 : 1).$$

Moreover, it contracts the coordinate lines:

$$Z(x) \text{ to } p_1, \quad Z(y) \text{ to } p_2, \quad Z(z) \text{ to } p_3.$$

The map is birational, and it is an involution as a rational map:

$$\sigma \circ \sigma = \text{id}.$$

Resolution by blow-ups

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of $\mathbb{P}^2$ at the three base points p_1, p_2, p_3 . Then σ lifts to a morphism

$$\tilde{\sigma} : \widetilde{\mathbb{P}^2} \rightarrow \widetilde{\mathbb{P}^2}.$$

In the Picard group of $\widetilde{\mathbb{P}^2}$, if H denotes the pullback of the class of a line and E_1, E_2, E_3 are the exceptional divisors, then

$$\sigma^*(H) = 2H - E_1 - E_2 - E_3,$$

$$\sigma^*(E_1) = H - E_2 - E_3, \quad \sigma^*(E_2) = H - E_1 - E_3, \quad \sigma^*(E_3) = H - E_1 - E_2.$$

Effect on degrees of plane curves

Let $C \subset \mathbb{P}^2$ be a curve of degree d . Assume C passes through the base points with multiplicities

$$m_1, \quad m_2, \quad m_3.$$

Then the strict transform of C under the Cremona involution has degree

$$d' = 2d - m_1 - m_2 - m_3.$$

For example, a general line has $d = 1$ and $m_1 = m_2 = m_3 = 0$, hence

$$d' = 2,$$

so it maps to a conic through the three base points. A conic through all three base points has $d = 2$ and $m_1 = m_2 = m_3 = 1$, hence

$$d' = 1,$$

so it maps to a line.

Remark on cubics

For a cubic $d = 3$, if it passes simply through p_1, p_2, p_3 , then

$$d' = 2 \cdot 3 - 1 - 1 - 1 = 3,$$

so it transforms to another cubic. Higher multiplicities at the base points can decrease the degree after transformation.

What is a cubic

A *cubic* is an algebraic set defined by a polynomial equation of total degree 3.

Affine case. A cubic hypersurface in $\mathbb{A}^n$ is

$$X = Z(f) \subset \mathbb{A}^n, \quad f \in k[x_1, \dots, x_n], \quad \deg(f) = 3.$$

Projective case. A cubic hypersurface in $\mathbb{P}^n$ is

$$X = Z(F) \subset \mathbb{P}^n, \quad F \in k[x_0, \dots, x_n] \text{ homogeneous}, \quad \deg(F) = 3.$$

Curve versus surface. A hypersurface in $\mathbb{A}^n$ (or $\mathbb{P}^n$) typically has dimension $n - 1$. Hence:

$$\begin{aligned} \text{cubic in } \mathbb{A}^2 \text{ or } \mathbb{P}^2 &\text{ is a curve,} \\ \text{cubic in } \mathbb{A}^3 \text{ or } \mathbb{P}^3 &\text{ is a surface.} \end{aligned}$$

Affine cubic versus projective cubic: are they the same?

Let k be a field.

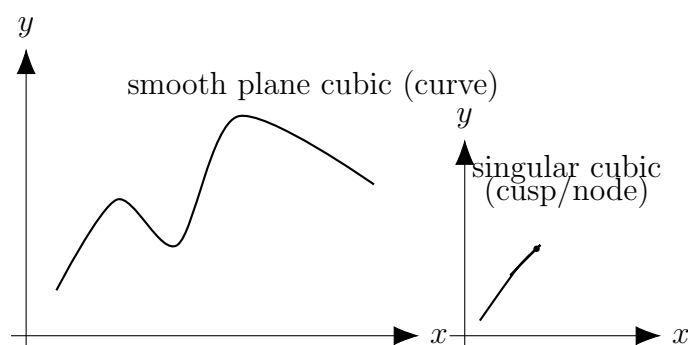


Figure 1: In $\mathbb{A}^2$ or $\mathbb{P}^2$, a cubic is typically a *curve*. Smooth cubics are elliptic curves; singular irreducible cubics are rational.

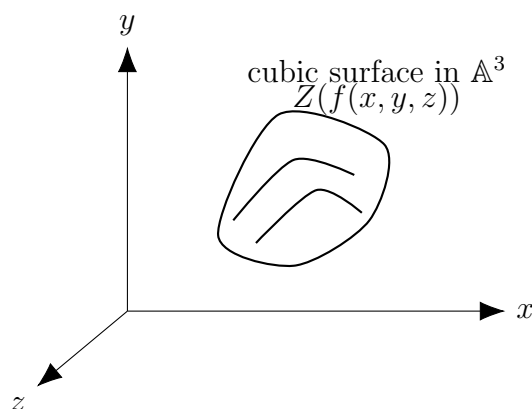


Figure 2: In $\mathbb{A}^3$ (or $\mathbb{P}^3$), a cubic is typically a *surface*: one degree-3 equation cuts out a 2D object.

Affine hypersurfaces

A *cubic affine hypersurface* in $\mathbb{A}^n$ is the zero locus of a polynomial of total degree at most 3:

$$X_{\text{aff}} = Z(f) \subset \mathbb{A}^n, \quad f \in k[x_1, \dots, x_n], \quad \deg(f) \leq 3.$$

Projective hypersurfaces

A *cubic projective hypersurface* in $\mathbb{P}^n$ is the zero locus of a homogeneous polynomial of degree 3:

$$X_{\text{proj}} = Z(F) \subset \mathbb{P}^n, \quad F \in k[x_0, x_1, \dots, x_n] \text{ homogeneous}, \quad \deg(F) = 3.$$

In projective space, equations must be homogeneous.

Homogenization and projective closure

Given $f \in k[x_1, \dots, x_n]$ with $\deg(f) \leq 3$, its homogenization of degree 3 is the homogeneous polynomial

$$F(x_0, x_1, \dots, x_n) = x_0^3 f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

Equivalently, each monomial

$$x_1^{a_1} \cdots x_n^{a_n}$$

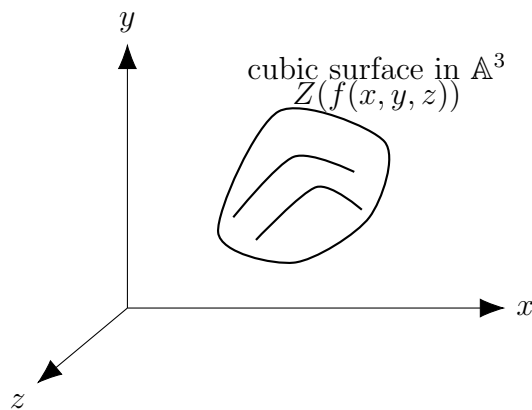


Figure 3: In $\mathbb{A}^3$ (or $\mathbb{P}^3$), a cubic is typically a *surface*: one degree-3 equation cuts out a 2D object.

of total degree

$$a_1 + \cdots + a_n = d \leq 3$$

is replaced by

$$x_1^{a_1} \cdots x_n^{a_n} x_0^{3-d}.$$

The projective hypersurface

$$X_{\text{proj}} = Z(F) \subset \mathbb{P}^n$$

is called the *projective closure* of

$$X_{\text{aff}} = Z(f) \subset \mathbb{A}^n.$$

Affine chart and points at infinity

Consider the standard affine chart

$$U_0 = \{x_0 \neq 0\} \subset \mathbb{P}^n,$$

with coordinates

$$x'_i = \frac{x_i}{x_0}.$$

On this chart we have

$$F(1, x_1, \dots, x_n) = f(x_1, \dots, x_n).$$

Hence

$$X_{\text{proj}} \cap U_0 \cong X_{\text{aff}}.$$

So the affine hypersurface is the part of the projective hypersurface lying in the chart $x_0 \neq 0$.

However, X_{proj} may contain *additional points at infinity*, namely

$$X_{\text{proj}} \cap \{x_0 = 0\}.$$

Thus, in general,

$$X_{\text{proj}} = \overline{X_{\text{aff}}}^{\text{proj}} = X_{\text{aff}} \cup (X_{\text{proj}} \cap \{x_0 = 0\}),$$

where $\overline{X_{\text{aff}}}^{\text{proj}}$ denotes the projective closure.

Example

In $\mathbb{A}^2$ consider the affine cubic curve

$$f(x, y) = y - x^3.$$

Homogenizing to degree 3 gives a projective cubic in $\mathbb{P}^2$:

$$F(x, y, z) = yz^2 - x^3.$$

On the affine chart $z = 1$,

$$F(x, y, 1) = y - x^3,$$

so this recovers the affine curve. At infinity $z = 0$,

$$F(x, y, 0) = -x^3,$$

so $x = 0$, and the projective closure contains the point

$$(0 : 1 : 0).$$

Hence the projective cubic is the affine cubic together with an extra point at infinity.

Why projective closures can be larger although homogeneous polynomials are more restrictive

Let k be a field.

Homogeneous polynomials are a smaller class

A polynomial

$$f \in k[x_1, \dots, x_n]$$

is called *homogeneous of degree d* if every monomial occurring in f has total degree d . For example, in $k[x, y, z]$ the polynomial

$$x^3 + x^2y + yz^2$$

is homogeneous of degree 3, whereas

$$x^3 + y$$

is not homogeneous, because the monomials have degrees 3 and 1.

Thus, as polynomials, homogeneous degree-3 polynomials form a *more restrictive* class than arbitrary degree-3 polynomials.

Why the geometric set becomes larger in projective space

The apparent paradox comes from comparing objects in *different ambient spaces*. Affine space $\mathbb{A}^n$ sits inside projective space $\mathbb{P}^n$ as the standard chart

$$U_0 = \{x_0 \neq 0\} \subset \mathbb{P}^n,$$

via

$$(x_1, \dots, x_n) \mapsto (1 : x_1 : \dots : x_n).$$

Projective space also contains the hyperplane at infinity

$$H_\infty = \{x_0 = 0\} \subset \mathbb{P}^n,$$

which has no affine counterpart.

Given an affine equation

$$f(x_1, \dots, x_n) = 0$$

one forms its homogenization F and considers the projective hypersurface

$$Z(F) \subset \mathbb{P}^n.$$

On the affine chart U_0 one recovers the original affine hypersurface:

$$Z(F) \cap U_0 \cong Z(f) \subset \mathbb{A}^n.$$

However, $Z(F)$ may also meet the hyperplane at infinity H_∞ . Hence, in general,

$$Z(F) = (Z(F) \cap U_0) \cup (Z(F) \cap H_\infty),$$

so $Z(F)$ can be larger than the affine set $Z(f)$ because it may contain additional points at infinity.

What points at infinity are added

Assume $\deg(f) \leq 3$ and write f as a sum of homogeneous parts:

$$f = f_3 + f_2 + f_1 + f_0,$$

where f_d is homogeneous of degree d .

The homogenization of degree 3 is

$$F(x_0, x_1, \dots, x_n) = f_3(x_1, \dots, x_n) + x_0 f_2(x_1, \dots, x_n) + x_0^2 f_1(x_1, \dots, x_n) + x_0^3 f_0.$$

Restricting to the hyperplane at infinity $H_\infty = \{x_0 = 0\}$ gives

$$F(0, x_1, \dots, x_n) = f_3(x_1, \dots, x_n).$$

Therefore the points at infinity of the projective closure are exactly the projective zeros of the top-degree part:

$$Z(F) \cap H_\infty = Z(f_3) \subset \mathbb{P}^{n-1}.$$

So the geometry at infinity is controlled by the leading homogeneous term of f .

Example

Consider the affine cubic curve in $\mathbb{A}^2$:

$$f(x, y) = y - x^3.$$

Its top-degree part is

$$f_3(x, y) = -x^3.$$

Homogenizing to degree 3 in $\mathbb{P}^2$ yields

$$F(x, y, z) = yz^2 - x^3.$$

On the affine chart $z = 1$,

$$F(x, y, 1) = y - x^3,$$

so $Z(F) \cap \{z \neq 0\}$ is the original affine curve.

At infinity $z = 0$,

$$F(x, y, 0) = -x^3,$$

so

$$x = 0,$$

and the projective closure contains the point

$$(0 : 1 : 0).$$

Thus the projective curve is larger than the affine curve by the additional point at infinity.

Summary

Homogeneous polynomials are more restrictive as equations, but projective space contains additional points (the hyperplane at infinity). When one passes from an affine hypersurface $Z(f) \subset \mathbb{A}^n$ to its projective closure $Z(F) \subset \mathbb{P}^n$, one typically adds solutions at infinity, which is why $Z(F)$ can be larger than $Z(f)$.

Picard group and blow-ups (plane case, with applications to Cremona)

Divisors and the Picard group

Let X be a smooth irreducible variety over an algebraically closed field k .

Weil divisors

A *Weil divisor* on X is a finite formal integer linear combination of irreducible codimension-1 subvarieties:

$$D = \sum_i n_i V_i, \quad n_i \in \mathbb{Z}.$$

The group of Weil divisors is denoted

$$\text{Div}(X).$$

A divisor is called *effective* if

$$n_i \geq 0$$

for all i , and we write

$$D \geq 0.$$

Principal divisors

If $f \in k(X)^\times$ is a nonzero rational function on X , then its divisor of zeros and poles is a divisor

$$\operatorname{div}(f) \in \operatorname{Div}(X),$$

called a *principal divisor*. Two divisors are *linearly equivalent* if their difference is principal:

$$D \sim D' \iff D - D' = \operatorname{div}(f) \text{ for some } f \in k(X)^\times.$$

Picard group

The *Picard group* of X is

$$\operatorname{Pic}(X) = \{\text{line bundles on } X\} / \cong$$

with group law given by tensor product.

On a smooth variety, one has the fundamental identification

$$\operatorname{Pic}(X) \cong \operatorname{Div}(X) / \sim,$$

so the Picard group may be viewed as divisors modulo linear equivalence.

Picard group of projective space

For projective space,

$$\operatorname{Pic}(\mathbb{P}^n) \cong \mathbb{Z}.$$

A generator is the class of a hyperplane divisor H . Thus every line bundle is of the form

$$\mathcal{O}_{\mathbb{P}^n}(d), \quad d \in \mathbb{Z},$$

and divisor classes are multiples of H :

$$\operatorname{Pic}(\mathbb{P}^n) = \mathbb{Z} \cdot H.$$

In particular, on $\mathbb{P}^2$ we write H for the class of a line.

Blow-up at a point: geometric idea

Let X be a smooth surface and let $p \in X$ be a point. The *blow-up of X at p* is a morphism

$$\pi : \tilde{X} \rightarrow X$$

such that:

1. π is an isomorphism over $X \setminus \{p\}$,
2. the fiber $\pi^{-1}(p)$ is a copy of $\mathbb{P}^1$, called the *exceptional divisor*.

Intuitively, blowing up replaces the point p by the space of directions through p . On a surface, directions are parametrized by $\mathbb{P}^1$, which explains why the exceptional fiber is $\mathbb{P}^1$.

Local model in affine coordinates

In local affine coordinates (u, v) near $p = (0, 0) \in \mathbb{A}^2$, the blow-up

$$\mathrm{Bl}_0(\mathbb{A}^2) \subset \mathbb{A}^2 \times \mathbb{P}^1$$

is given by the incidence relation

$$ub = va$$

in coordinates $(u, v) \in \mathbb{A}^2$ and $(a : b) \in \mathbb{P}^1$. The projection

$$\pi(u, v, (a : b)) = (u, v)$$

is the blow-down map. The exceptional divisor is

$$E = \pi^{-1}(0, 0) \cong \mathbb{P}^1.$$

Picard group under blow-up of $\mathbb{P}^2$

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of $\mathbb{P}^2$ at one point p . Let E be the exceptional divisor.

Generators and structure

The Picard group becomes rank 2:

$$\mathrm{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E,$$

where H denotes the pullback of the class of a line:

$$H = \pi^*(\text{line class}).$$

If we blow up r distinct points $p_1, \dots, p_r$, then

$$\mathrm{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E_1 \oplus \dots \oplus \mathbb{Z} \cdot E_r.$$

Intersection numbers

On a smooth surface, divisor classes have an intersection product. For the blow-up of $\mathbb{P}^2$ at r points one has

$$H^2 = 1, \quad H \cdot E_i = 0, \quad E_i \cdot E_j = 0 \text{ for } i \neq j, \quad E_i^2 = -1.$$

The key geometric meaning is:

- $H^2 = 1$ because two general lines meet in one point.
- $E_i^2 = -1$ reflects that E_i is an exceptional curve of the first kind.

Strict transforms and multiplicities

Let $C \subset \mathbb{P}^2$ be a plane curve of degree d . Assume C passes through p_i with multiplicity m_i . After blowing up at the points p_i , the class of the strict transform $\tilde{C}$ in the Picard group is

$$[\tilde{C}] = dH - \sum_{i=1}^r m_i E_i.$$

This formula records two pieces of data:

1. the degree d of the original curve,
2. how much of C is concentrated at the blown-up points, via the multiplicities m_i .

Cremona involution via blow-ups (why Picard matters)

Let

$$\sigma : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$$

be the standard Cremona transformation

$$(x : y : z) \mapsto (yz : zx : xy),$$

with base points

$$p_1 = (1 : 0 : 0), \quad p_2 = (0 : 1 : 0), \quad p_3 = (0 : 0 : 1).$$

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up at p_1, p_2, p_3 , with exceptional divisors E_1, E_2, E_3 and line class H .

Then σ lifts to a morphism on $\widetilde{\mathbb{P}^2}$, and its action on divisor classes is encoded by

$$\sigma^*(H) = 2H - E_1 - E_2 - E_3,$$

$$\sigma^*(E_1) = H - E_2 - E_3, \quad \sigma^*(E_2) = H - E_1 - E_3, \quad \sigma^*(E_3) = H - E_1 - E_2.$$

Degree change formula

Let $C \subset \mathbb{P}^2$ be a curve of degree d , with multiplicities m_1, m_2, m_3 at the base points. The strict transform of $\sigma(C)$ has degree

$$d' = 2d - m_1 - m_2 - m_3.$$

This is obtained by tracking the class

$$[\tilde{C}] = dH - m_1 E_1 - m_2 E_2 - m_3 E_3$$

under the pullback action on $\text{Pic}(\widetilde{\mathbb{P}^2})$.

Example

A general line has $d = 1$ and $m_1 = m_2 = m_3 = 0$, hence

$$d' = 2,$$

so it maps to a conic. A conic through all three base points has $d = 2$ and $m_1 = m_2 = m_3 = 1$, hence

$$d' = 1,$$

so it maps to a line.

Picard group and blow-ups (plane case)**Divisors and the Picard group**

Let X be a smooth irreducible variety over an algebraically closed field k .

Weil divisors

A *Weil divisor* on X is a finite formal integer linear combination of irreducible codimension-1 subvarieties:

$$D = \sum_i n_i V_i, \quad n_i \in \mathbb{Z}.$$

The group of Weil divisors is denoted

$$\text{Div}(X).$$

Principal divisors and linear equivalence

If $f \in k(X)^\times$ is a nonzero rational function on X , then f defines a divisor of zeros and poles

$$\text{div}(f) \in \text{Div}(X),$$

called a *principal divisor*. Two divisors are *linearly equivalent* if their difference is principal:

$$D \sim D' \iff D - D' = \text{div}(f) \text{ for some } f \in k(X)^\times.$$

Picard group

The *Picard group* of X is the group of line bundles on X up to isomorphism:

$$\text{Pic}(X) = \{\text{line bundles on } X\} / \cong,$$

with group law given by tensor product. On a smooth variety there is an identification

$$\text{Pic}(X) \cong \text{Div}(X) / \sim,$$

so $\text{Pic}(X)$ can be viewed as divisor classes modulo linear equivalence.

Picard group of $\mathbb{P}^2$

For projective space one has

$$\mathrm{Pic}(\mathbb{P}^n) \cong \mathbb{Z}.$$

A generator is the hyperplane class H . In particular,

$$\mathrm{Pic}(\mathbb{P}^2) = \mathbb{Z} \cdot H,$$

and every divisor class on $\mathbb{P}^2$ is of the form dH .

0.1 Blow-up of a point on a surface

Let X be a smooth surface and let $p \in X$. The *blow-up of X at p* is a morphism

$$\pi : \tilde{X} \rightarrow X$$

such that:

1. π is an isomorphism over $X \setminus \{p\}$,
2. the fiber $\pi^{-1}(p)$ is a curve

$$E = \pi^{-1}(p) \cong \mathbb{P}^1,$$

called the *exceptional divisor*.

Geometrically, blowing up replaces the point p by the set of tangent directions through p .

0.1.1 Local model in affine coordinates

In local affine coordinates (u, v) near $p = (0, 0) \in \mathbb{A}^2$, the blow-up

$$\mathrm{Bl}_0(\mathbb{A}^2) \subset \mathbb{A}^2 \times \mathbb{P}^1$$

is given by

$$ub = va$$

in coordinates $(u, v) \in \mathbb{A}^2$ and $(a : b) \in \mathbb{P}^1$. The projection

$$\pi(u, v, (a : b)) = (u, v)$$

is the blow-down map, and the exceptional divisor is $E = \{u = v = 0\} \cong \mathbb{P}^1$.

Picard group after blowing up $\mathbb{P}^2$

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of $\mathbb{P}^2$ at r distinct points $p_1, \dots, p_r$, with exceptional divisors $E_1, \dots, E_r$. Let H denote the pullback of the class of a line.

Then

$$\mathrm{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E_1 \oplus \dots \oplus \mathbb{Z} \cdot E_r.$$

Intersection numbers

On the blow-up of $\mathbb{P}^2$ at r points, the intersection product satisfies

$$H^2 = 1, \quad H \cdot E_i = 0, \quad E_i \cdot E_j = 0 \text{ for } i \neq j, \quad E_i^2 = -1.$$

Strict transforms and multiplicities

Let $C \subset \mathbb{P}^2$ be a plane curve of degree d . Assume C passes through p_i with multiplicity m_i . Then the divisor class of the strict transform $\tilde{C} \subset \widetilde{\mathbb{P}^2}$ is

$$[\tilde{C}] = dH - \sum_{i=1}^r m_i E_i.$$

Remark: connection to Cremona

For the standard Cremona involution with base points p_1, p_2, p_3 , one blows up these three points to obtain

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2.$$

The Cremona map lifts to a morphism on $\widetilde{\mathbb{P}^2}$ and acts linearly on

$$\text{Pic}(\widetilde{\mathbb{P}^2}).$$

Tracking a curve class

$$dH - m_1 E_1 - m_2 E_2 - m_3 E_3$$

under this action yields the degree change formula

$$d' = 2d - m_1 - m_2 - m_3.$$

Charts for blow-ups, exceptional divisor, and Cremona

Chart 1: Blow-up of a point on a surface

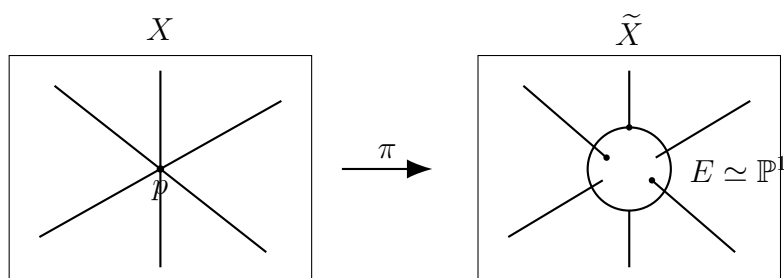


Figure 4: Blowing up replaces the point p by the exceptional divisor $E \simeq \mathbb{P}^1$. Different directions through p become different points on E .

Chart 2: Picard group generators and intersection form on $\text{Bl}_p(\mathbb{P}^2)$

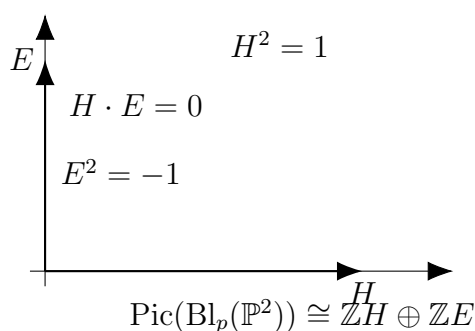


Figure 5: After blowing up one point, the Picard group gains the new generator E (the exceptional divisor), with intersection numbers $H^2 = 1$, $H \cdot E = 0$, and $E^2 = -1$.

Chart 3: Cremona involution and the three base points

Concrete divisors, codimension, and line bundles

Why divisors use codimension one

Let X be an irreducible variety. A *prime divisor* on X means an irreducible closed subvariety of codimension 1. A *Weil divisor* is a finite formal integer combination of prime divisors:

$$D = \sum_i n_i V_i, \quad n_i \in \mathbb{Z}.$$

The reason codimension 1 appears is that zeros and poles of a rational function occur along hypersurfaces. For a nonzero rational function $f \in k(X)^\times$ one defines the *principal divisor*

$$\text{div}(f) = \sum_V \text{ord}_V(f) V,$$

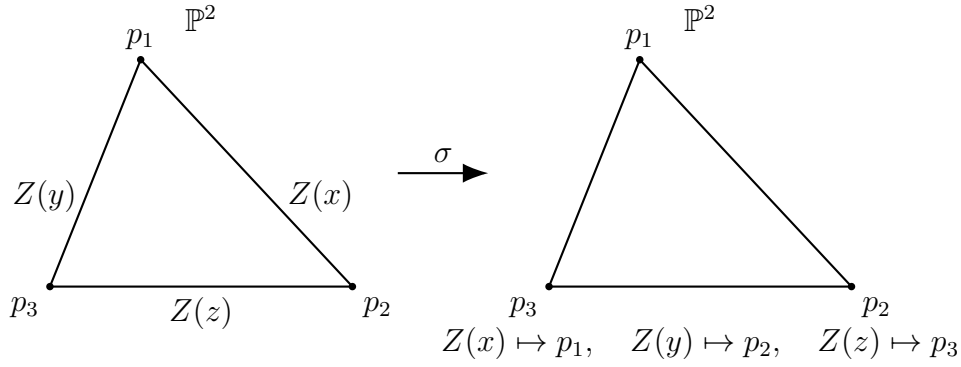


Figure 6: The standard Cremona map has base points p_1, p_2, p_3 and contracts the coordinate lines to these points. After blowing up the base points, the map becomes regular on the blow-up.

where the sum runs over prime divisors $V \subset X$ and $\text{ord}_V(f)$ is the order of vanishing (negative for poles) along V . Higher codimension sets do not contribute to $\text{div}(f)$.

In particular:

$$\dim(X) = 1 \implies \text{prime divisors are points,}$$

$$\dim(X) = 2 \implies \text{prime divisors are curves.}$$

Example: divisors on $\mathbb{P}^1$

On $X = \mathbb{P}^1$, prime divisors are points. Using the affine coordinate $t = x/y$, consider

$$f(t) = \frac{t-a}{t-b} \in k(t)^\times.$$

Then f has a simple zero at $t = a$ and a simple pole at $t = b$, hence

$$\text{div}(f) = (a) - (b).$$

One has

$$\text{Pic}(\mathbb{P}^1) \cong \mathbb{Z},$$

and the degree of a divisor

$$D = \sum_i n_i(p_i)$$

is

$$\deg(D) = \sum_i n_i.$$

Every principal divisor has degree 0.

Example: divisors on $\mathbb{P}^2$

On $X = \mathbb{P}^2$, prime divisors are irreducible plane curves. Let $(x : y : z)$ be homogeneous coordinates and consider the rational function

$$f = \frac{x}{z} \in k(\mathbb{P}^2)^\times.$$

The zero locus of f is the line $Z(x)$ and the pole locus is the line $Z(z)$, so

$$\operatorname{div}\left(\frac{x}{z}\right) = Z(x) - Z(z).$$

Thus $Z(x) \sim Z(z)$, and in fact all lines are linearly equivalent. One has

$$\operatorname{Pic}(\mathbb{P}^2) \cong \mathbb{Z},$$

generated by the class H of a line.

Example: blow-up of $\mathbb{P}^2$ at a point

Let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of $\mathbb{P}^2$ at a point p . The exceptional divisor is

$$E = \pi^{-1}(p) \cong \mathbb{P}^1.$$

The Picard group becomes

$$\operatorname{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E,$$

where H is the pullback of the class of a line.

If $C \subset \mathbb{P}^2$ is a curve of degree d and has multiplicity m at p , then the class of its strict transform satisfies

$$[\widetilde{C}] = dH - mE.$$

For example, a line not passing through p has strict transform class H , while a line passing through p has strict transform class $H - E$.

Line bundles are not projective lines

A *line bundle* on X is a rank-1 vector bundle. Equivalently, it is a morphism

$$\pi : L \rightarrow X$$

such that for each $x \in X$ the fiber L_x is a 1-dimensional k -vector space and locally

$$L|_U \cong U \times k.$$

Thus a line bundle is a family of 1-dimensional vector spaces parametrized by X .

The variety $\mathbb{P}^1$ is a projective line, not a line bundle. A $\mathbb{P}^1$ -bundle arises as the projectivization $\mathbb{P}(E)$ of a rank-2 vector bundle E . If L is a line bundle (rank 1), then its projectivization has fibers $\mathbb{P}^0$ (a point), so

$$\mathbb{P}(L) \cong X.$$

One blow-up of $\mathbb{P}^2$: concrete divisor classes and computations

Let $p \in \mathbb{P}^2$ be a point and let

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

be the blow-up of $\mathbb{P}^2$ at p . Let

$$E = \pi^{-1}(p) \cong \mathbb{P}^1$$

be the exceptional divisor. Let H denote the pullback of the class of a line in $\mathbb{P}^2$.

Picard group and intersection pairing

One has

$$\mathrm{Pic}(\widetilde{\mathbb{P}^2}) \cong \mathbb{Z} \cdot H \oplus \mathbb{Z} \cdot E,$$

and the intersection numbers satisfy

$$H^2 = 1, \quad H \cdot E = 0, \quad E^2 = -1.$$

For divisor classes

$$D = aH - bE, \quad D' = cH - dE,$$

the intersection product is

$$D \cdot D' = ac - bd.$$

Strict transform class, degree, and multiplicity

Let $C \subset \mathbb{P}^2$ be a plane curve of degree d , and assume C has multiplicity m at p . Then the divisor class of the strict transform $\widetilde{C} \subset \widetilde{\mathbb{P}^2}$ is

$$[\widetilde{C}] = dH - mE.$$

Moreover, degree and multiplicity can be recovered as intersection numbers:

$$\widetilde{C} \cdot H = d, \quad \widetilde{C} \cdot E = m.$$

Indeed,

$$(dH - mE) \cdot H = dH^2 - m(E \cdot H) = d,$$

and

$$(dH - mE) \cdot E = d(H \cdot E) - mE^2 = m.$$

Examples

1. A line L not passing through p has strict transform class

$$[\widetilde{L}] = H,$$

and

$$\widetilde{L}^2 = H^2 = 1.$$

2. A line L passing through p has multiplicity $m = 1$ at p , hence

$$[\widetilde{L}] = H - E.$$

Its self-intersection is

$$(H - E)^2 = H^2 - 2(H \cdot E) + E^2 = 1 - 1 = 0,$$

and it meets the exceptional divisor once:

$$(H - E) \cdot E = 1.$$

3. A conic Q not passing through p has class

$$[\tilde{Q}] = 2H,$$

so

$$\tilde{Q}^2 = (2H)^2 = 4.$$

4. A conic Q passing through p simply has multiplicity $m = 1$, hence

$$[\tilde{Q}] = 2H - E.$$

Then

$$(2H - E)^2 = 4 - 1 = 3, \quad (2H - E) \cdot E = 1.$$

5. A conic Q with multiplicity $m = 2$ at p has class

$$[\tilde{Q}] = 2H - 2E,$$

so

$$(2H - 2E)^2 = 4 - 4 = 0, \quad (2H - 2E) \cdot E = 2.$$

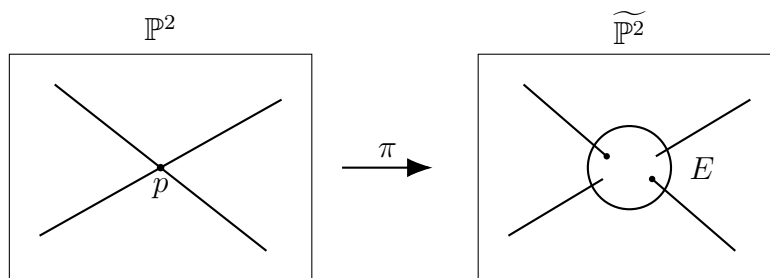


Figure 7: Blowing up replaces p by $E \cong \mathbb{P}^1$; different directions through p become different points on E .

The tautological line bundle on projective space

Let V be a finite-dimensional k -vector space, $\dim(V) = n + 1$. Projective space $\mathbb{P}(V)$ is the set of 1-dimensional linear subspaces $\ell \subset V$. A point of $\mathbb{P}(V)$ is written $[\ell]$.

The tautological bundle $\mathcal{O}_{\mathbb{P}(V)}(-1)$

There is a canonical rank-1 subbundle of the trivial bundle $\mathbb{P}(V) \times V$:

$$\mathcal{O}_{\mathbb{P}(V)}(-1) \subset \mathbb{P}(V) \times V,$$

defined by

$$\mathcal{O}_{\mathbb{P}(V)}(-1) = \{ ([\ell], v) \in \mathbb{P}(V) \times V \mid v \in \ell \}.$$

Its fiber over a point $[\ell] \in \mathbb{P}(V)$ is the line itself:

$$\mathcal{O}_{\mathbb{P}(V)}(-1)_{[\ell]} = \ell.$$

This is why it is called the tautological line bundle.

The principal $k^\times$ -bundle and the relation to $\mathcal{O}(-1)$

There is a natural quotient map

$$V \setminus \{0\} \longrightarrow \mathbb{P}(V), \quad v \longmapsto [kv],$$

whose fibers are $\ell \setminus \{0\}$. After choosing a basis of ℓ , one identifies $\ell \setminus \{0\}$ with $k^\times$. This makes $V \setminus \{0\} \rightarrow \mathbb{P}(V)$ a principal $k^\times$ -bundle. The line bundle $\mathcal{O}_{\mathbb{P}(V)}(-1)$ can be viewed as the associated line bundle obtained by adjoining the zero vector in each fiber, but the intrinsic definition above is standard.

The hyperplane bundle $\mathcal{O}(1)$ and the Picard group

Define the dual line bundle

$$\mathcal{O}_{\mathbb{P}(V)}(1) = \mathcal{O}_{\mathbb{P}(V)}(-1)^\vee.$$

On $\mathbb{P}^n = \mathbb{P}(k^{n+1})$, one has

$$\text{Pic}(\mathbb{P}^n) \cong \mathbb{Z},$$

and the class of $\mathcal{O}(1)$ is a generator. Hyperplanes are precisely the zero loci of global sections of $\mathcal{O}(1)$, and more generally

$$\mathcal{O}(d) = \mathcal{O}(1)^{\otimes d}$$

corresponds to degree- d hypersurfaces.

Local trivializations on standard charts

Let $[x_0 : \cdots : x_n]$ be homogeneous coordinates on $\mathbb{P}^n$ and set

$$U_i = \{x_i \neq 0\}.$$

On each U_i , the bundle $\mathcal{O}(-1)$ is trivial. On overlaps $U_i \cap U_j$, the transition function for $\mathcal{O}(-1)$ is

$$g_{ij} = \frac{x_j}{x_i}.$$

For the dual bundle $\mathcal{O}(1)$ the transition function is

$$g_{ij}^{-1} = \frac{x_i}{x_j}.$$

Connection to blow-ups

If

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

is the blow-up at a point, then the line class H used in Picard computations on $\widetilde{\mathbb{P}^2}$ is the pullback of the hyperplane bundle:

$$H = \pi^* \mathcal{O}_{\mathbb{P}^2}(1).$$

From divisors to line bundles: $\mathcal{O}(D)$ and zero loci of sections

Let X be a smooth irreducible variety over an algebraically closed field k . On a smooth variety, divisor classes and line bundles are closely linked.

Cartier divisors and local equations

A Cartier divisor D on X can be described by an open cover $\{U_i\}$ and rational functions

$$f_i \in k(X)^\times$$

such that on overlaps $U_i \cap U_j$ one has

$$f_i/f_j \in \mathcal{O}_X^\times(U_i \cap U_j).$$

Intuitively, each f_i is a local defining equation for the divisor on U_i , and the compatibility says that two local equations differ by a nowhere-vanishing regular function.

On a smooth variety every Weil divisor is Cartier, so the construction below applies to Weil divisors as well.

Definition of the line bundle $\mathcal{O}(D)$

Given a Cartier divisor D represented by local equations $\{(U_i, f_i)\}$, define a line bundle $\mathcal{O}(D)$ by gluing trivial bundles on each U_i with transition functions

$$g_{ij} = \frac{f_i}{f_j} \in \mathcal{O}_X^\times(U_i \cap U_j).$$

Equivalently, one may define the associated invertible sheaf by

$$\mathcal{O}(D)(U) = \{h \in k(X) \mid hf_i \in \mathcal{O}_X(U \cap U_i) \text{ for all } i\}.$$

This construction depends only on the linear equivalence class of D .

Linear equivalence corresponds to isomorphic line bundles

If

$$D \sim D',$$

then

$$D - D' = \text{div}(\varphi)$$

for some $\varphi \in k(X)^\times$. In this case one has a canonical isomorphism of line bundles

$$\mathcal{O}(D) \cong \mathcal{O}(D').$$

Hence the assignment $D \mapsto \mathcal{O}(D)$ factors through divisor classes and gives the identification

$$\text{Pic}(X) \cong \text{Div}(X)/\sim.$$

Sections and divisors of zeros and poles

A rational section s of $\mathcal{O}(D)$ can be viewed locally as a rational function multiplied by the local trivializing section. Its divisor of zeros and poles, denoted $\text{div}(s)$, is a Weil divisor. There is a fundamental relation:

$$\text{div}(s) + D \geq 0 \iff s \text{ is a regular section of } \mathcal{O}(D).$$

In particular, if s is a nonzero regular section of $\mathcal{O}(D)$, then its zero locus defines an effective divisor and one has

$$Z(s) = \text{div}(s) + D.$$

Thus divisors can be realized as zero loci of sections of associated line bundles.

Example: $\mathbb{P}^n$, hyperplanes, and $\mathcal{O}(1)$

Let $H \subset \mathbb{P}^n$ be a hyperplane. Then the associated line bundle is the hyperplane bundle:

$$\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1).$$

A hyperplane $H = Z(L)$ is given by a nonzero linear form L , and L defines a global section of $\mathcal{O}(1)$ whose zero locus is exactly H :

$$Z(L) = H.$$

More generally, for $d \geq 1$,

$$\mathcal{O}_{\mathbb{P}^n}(d) = \mathcal{O}_{\mathbb{P}^n}(1)^{\otimes d},$$

and homogeneous polynomials of degree d define global sections of $\mathcal{O}(d)$ whose zero loci are degree- d hypersurfaces.

Connection to blow-ups

If

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

is the blow-up at a point, then the class H used in divisor computations on $\widetilde{\mathbb{P}^2}$ is the pullback of the hyperplane bundle:

$$\mathcal{O}_{\widetilde{\mathbb{P}^2}}(H) \cong \pi^* \mathcal{O}_{\mathbb{P}^2}(1).$$

Projective bundles $\mathbb{P}(E) \rightarrow X$

Let X be a variety and let

$$p : E \rightarrow X$$

be a vector bundle (locally free sheaf) of rank r . For each point $x \in X$, the fiber E_x is an r -dimensional k -vector space.

Definition of the projective bundle

The *projective bundle* associated to E is a morphism

$$\pi : \mathbb{P}(E) \rightarrow X$$

whose fiber over x is the projective space of the fiber:

$$\pi^{-1}(x) \cong \mathbb{P}(E_x) \cong \mathbb{P}^{r-1}.$$

Thus:

$$\begin{aligned} \text{rank}(E) = 2 &\implies \mathbb{P}(E) \rightarrow X \text{ is a } \mathbb{P}^1\text{-bundle,} \\ \text{rank}(E) = 1 &\implies \mathbb{P}(E) \rightarrow X \text{ has fibers } \mathbb{P}^0 \text{ and } \mathbb{P}(E) \cong X. \end{aligned}$$

Local triviality

Projective bundles are locally trivial in the Zariski topology. If E is trivial on an open set $U \subset X$, meaning

$$E|_U \cong U \times k^r,$$

then

$$\mathbb{P}(E)|_U \cong U \times \mathbb{P}^{r-1}.$$

Tautological line bundle and the universal exact sequence

There are two common conventions:

- $\mathbb{P}(E)$ parameterizes lines in the fibers E_x ,
- $\mathbb{P}(E)$ parameterizes 1-dimensional quotients of the fibers E_x .

Using the quotient convention, there is a canonical surjection on $\mathbb{P}(E)$ producing the *universal quotient line bundle*

$$\pi^*E \twoheadrightarrow \mathcal{O}_{\mathbb{P}(E)}(1),$$

and a short exact sequence

$$0 \rightarrow \mathcal{S} \rightarrow \pi^*E \rightarrow \mathcal{O}_{\mathbb{P}(E)}(1) \rightarrow 0,$$

where $\mathcal{S}$ is a rank $r - 1$ subbundle. The dual line bundle

$$\mathcal{O}_{\mathbb{P}(E)}(-1) = \mathcal{O}_{\mathbb{P}(E)}(1)^\vee$$

is often called the tautological line bundle.

Picard group of a projective bundle

Under mild hypotheses (for example X integral and E locally free), one has an isomorphism

$$\mathrm{Pic}(\mathbb{P}(E)) \cong \mathrm{Pic}(X) \oplus \mathbb{Z} \cdot [\mathcal{O}_{\mathbb{P}(E)}(1)].$$

Concretely, every line bundle on $\mathbb{P}(E)$ can be written uniquely as

$$\pi^*L \otimes \mathcal{O}_{\mathbb{P}(E)}(m), \quad L \in \mathrm{Pic}(X), \quad m \in \mathbb{Z}.$$

Example: blow-up exceptional divisor

If

$$\pi : \tilde{X} \rightarrow X$$

is the blow-up of a smooth variety X at a point p , then the exceptional divisor is a projective space:

$$E = \pi^{-1}(p) \cong \mathbb{P}^{\dim(X)-1}.$$

In particular, for the blow-up of $\mathbb{P}^2$ at a point one has

$$E \cong \mathbb{P}^1.$$

Geometrically, E parameterizes tangent directions at the blown-up point.

Concrete examples: sections of $\mathcal{O}(1)$ and $\mathcal{O}(d)$

Let $[x_0 : \cdots : x_n]$ be homogeneous coordinates on $\mathbb{P}^n$.

Hyperplanes as zero loci of sections of $\mathcal{O}(1)$

Let

$$L = a_0x_0 + \cdots + a_nx_n$$

be a nonzero linear form. The hyperplane is

$$H = Z(L) \subset \mathbb{P}^n.$$

The polynomial L defines a global section of $\mathcal{O}_{\mathbb{P}^n}(1)$, and its vanishing locus as a section is exactly H :

$$Z(L) = H.$$

On the standard chart

$$U_i = \{x_i \neq 0\},$$

the section is represented by the regular function

$$\frac{L}{x_i}.$$

Hence $H \cap U_i$ is cut out by one equation

$$\frac{L}{x_i} = 0$$

in the affine coordinates on U_i .

Degree- d hypersurfaces as zero loci of sections of $\mathcal{O}(d)$

Let

$$F \in k[x_0, \dots, x_n]$$

be homogeneous of degree $d \geq 1$. Then F defines a global section of

$$\mathcal{O}_{\mathbb{P}^n}(d), \quad \mathcal{O}_{\mathbb{P}^n}(d) = \mathcal{O}_{\mathbb{P}^n}(1)^{\otimes d},$$

and its zero locus is the degree- d hypersurface

$$Z(F) \subset \mathbb{P}^n.$$

On the chart U_i the section is represented by the regular function

$$\frac{F}{x_i^d}.$$

A principal divisor: why hyperplanes are linearly equivalent

Consider the rational function on $\mathbb{P}^n$

$$f = \frac{x_0}{x_1}.$$

Its divisor of zeros and poles is

$$\operatorname{div}(f) = Z(x_0) - Z(x_1).$$

Hence

$$Z(x_0) \sim Z(x_1),$$

so all hyperplanes have the same divisor class in $\operatorname{Pic}(\mathbb{P}^n)$, corresponding to $\mathcal{O}(1)$.

Concrete projective bundle example via polynomials: blow-up of $\mathbb{P}^2$ at a point

Let

$$p = (0 : 0 : 1) \in \mathbb{P}^2.$$

The blow-up of $\mathbb{P}^2$ at p can be realized as a closed subvariety of $\mathbb{P}^2 \times \mathbb{P}^1$:

$$\widetilde{\mathbb{P}^2} = \{((x : y : z), (s : t)) \in \mathbb{P}^2 \times \mathbb{P}^1 \mid xt - ys = 0\}.$$

The projection

$$\pi : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^2$$

is the blow-down map. The exceptional divisor is

$$E = \pi^{-1}(p) = \{((0 : 0 : 1), (s : t))\} \cong \mathbb{P}^1.$$

The second projection

$$\rho : \widetilde{\mathbb{P}^2} \rightarrow \mathbb{P}^1, \quad ((x : y : z), (s : t)) \longmapsto (s : t),$$

exhibits $\widetilde{\mathbb{P}^2}$ as a $\mathbb{P}^1$ -bundle over $\mathbb{P}^1$. Indeed, for fixed $(s : t)$ the fiber is

$$\rho^{-1}(s : t) = \{(x : y : z) \in \mathbb{P}^2 \mid tx - sy = 0\},$$

which is a projective line in $\mathbb{P}^2$, hence isomorphic to $\mathbb{P}^1$.